GLA University, Mathura
Department of Computer Engineering and Applications
Subject: Computer Organization
Subject Code: BCSC1005
Submission deadline: **September 25, 2025**

Assignment 1.1 (Module I)

1. A software developer is designing a new program that requires frequent access to both instructions and data during its execution.
How does the Von Neumann architecture facilitate the seamless integration of instructions and data in a way that enables efficient program execution? Provide specific examples to illustrate its advantage in programming.

2. Determine the floating-point numbers from the following Hexadecimal representation:

(i) 419C0000 H

(ii) C059700000000000 H

3. Suppose a robotics engineer is working on programming a robot's movement. The robot can move forward and backward in a grid pattern depending on the control commands. The robot's initial position is recorded at the 35th grid in an 8-bit processor. Compute the net movement of the robot if it follows the following commands.

Command 1: Move the robot forward by +45 units.
Command 2: Move the robot backward by -92 units.

4. An electrical engineer is designing a data acquisition system for a precision measurement application. The system requires representing a floating-point number in both single and double precision using the IEEE 754 standard.
In the data acquisition system, the engineer needs to convert the decimal number -307.1875 & +75.213 into IEEE 754 single and double precision hexadecimal notation. Perform the conversion and provide the resulting hexadecimal representations for both single and double precision formats.

5. Differentiate

i. Multiplexer and Demultiplexer

ii. Binary adder and CLA

iii. Memory read and Memory write

iv. Full Adder and Full Subtractor

The following transfer statements specify a memory. Explain the memory operation in each case.

i. ***R2 ← M[AR]***

ii. ***M[AR] ← R3***

iii. ***R5 ← M[R5]***

6. Imagine a customer adds multiple items to their shopping cart. How would you use a stack to manage the cart items efficiently, ensuring that the last item added is the first one removed when the customer proceeds to checkout? Explain PUSH and POP operations. Convert the following arithmetic expression into reverse polish notation (RPN) and show the stack operations for evaluating it.

a) 7 * 5 + (5 + 6) + (4 * 6 + 2)

b) 5 + 10 * 6 + (3 + 5) + 7 * (6 + 2)

c) 8 * 3 + (4 + 8) + (4 * 7 + 2) + 2 + 3

d) (A + B) * C - (D - E) * (F + G)

e) (((110/ (30-8))*50)+5)-((15/(10-5))*2)

f) $\frac{(10+3)[2*3+(3*9/3)]}{6-3+(4*3)}$

7. A computer architecture researcher is investigating the efficiency of different multiplication algorithms for signed numbers. In the research study, the researcher wants to explore the performance of Booth's algorithm for the multiplication of two signed numbers represented in 2's complement. Multiply -9 and -13 using Booth's algorithm, and store the result in the register.

8. Suppose you are building a basic microprocessor with eight 4-bit registers (labeled R0 through R7) that need to share data. Design a circuit using a common 4-bit bus, where a 3-bit select signal determines which register's binary content is transferred onto the bus. Include components like multiplexer and label all parts clearly.

9. Write 14-bit control word for the following microoperations:

(i) R1 ← R2 + R3

(ii) R4 ← R4

(iii) R5 ← R5 – 1

(iv) R6 ← Shl R1

(v) R7 ← Input

10. Suppose you're designing the arithmetic logic unit (ALU) for a simple 4-bit microprocessor that processes two binary numbers, **A ($A_3A_2A_1A_0$)** and **B ($B_3B_2B_1B_0$)**, to compute their sum or difference as S ($S_3S_2S_1S_0$). A control signal M determines the operation: when M is 0, it adds the numbers; when M is 1, it subtracts B from A. Additionally, include logic to detect and signal an overflow condition. Design the full combinational circuit with proper labelling.

11. Using general register computer with **3** address instruction, general register computer with **2** address instruction, accumulator type computer with **1** address instruction and stack organization computer with **zero** address instruction, write programs to evaluate the following arithmetic statements.

(i) A-B+(C+D)/E+F/G+(J*K)

(ii) (A*B+C)+D/E+F*G+I*J

12. Starting from an initial value of R = 11011101, determine the sequence of binary values in R after a logical shift-left, followed by a circular shift-right, followed by a logical shift-right and circular shift-left.

13. An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contains the number 200. Evaluate the effective address (EA) if the addressing mode of the instruction is: **Direct**, **Immediate**, **Relative**, **Register Indirect**, **Index with R1 as the index register**.

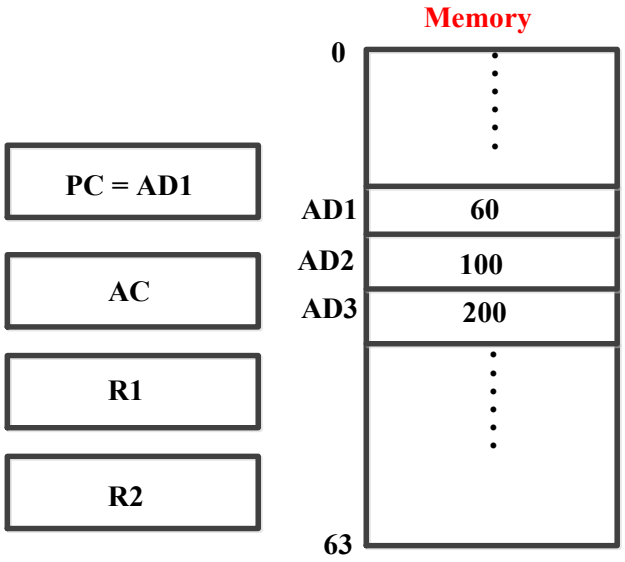
14. Considering the following organization of the 64 Byte memory and some processor registers, write the addressing mode of each of the following register transfer after executing the instructions:

(i) **R1 ← M [AD1] +M [AD2]**

(ii) **AC ← M [PC + AD3]**

(iii) **M [AD3] ← R2**

(iv) M [AD1] ← AC



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Assignment 1.2 (Module I)

1. A software developer is designing a new program that requires frequent access to both instructions and data during its execution.

How does the Von Neumann architecture facilitate the seamless integration of instructions and data in a way that enables efficient program execution? Provide specific examples to illustrate its advantage in programming.

2. Considering a 2's complement based 8-bit processor, perform the following arithmetic operations. Also, check the existence of overflow in each operation.

(a) (-100) + (15) (b) (-32) + (-97)

3. An electrical engineer is designing a data acquisition system for a precision measurement application. The system requires representing a floating-point number in both single and double precision using the IEEE 754 standard.

In the data acquisition system, the engineer needs to convert the decimal number -87.34 &
+78.213 into IEEE 754 single and double precision hexadecimal notation. Perform the conversion and provide the resulting hexadecimal representations for both single and double precision formats.

4. Find out the floating-point number from the following representation.

0 10000011 00111000000000000000000

5. You are provided with two half subtractor circuits. Describe how you will connect these half subtractors to design a full subtractor that can handle two binary digits and a borrow-in bit. Explain how the difference and borrow outputs are obtained using the logic gates involved.

6. The propagation time for carry and sum output of an adder is 20 ns and 30 ns, respectively. Calculate the maximum time delay in a 4-bit binary adder. Describe an efficient method to reduce this time latency.

7. Explain how CLA adder is the successor of the Ripple Carry Adder with logic diagram.

8. Design a combinational logic to add and subtract two 4-bit binary numbers. Also, design a logic to indicate the overflow condition.

9. Assume there are three wires red, green, and blue. One LED bulb is wired so that it will only illuminate when both the red and blue wires are experiencing high voltage at the same time. For

this system, utilize a 4x1 multiplexer to implement the digital circuit.

10. A computer architecture researcher is investigating the efficiency of different multiplication algorithms for signed numbers. In the research study, the researcher wants to explore the performance of Booth's algorithm for the multiplication of two signed numbers represented in 2's complement. Multiply -7 and -7 using Booth's algorithm, and store the result in the register.

11. Suppose you need to design a micro-operation to transfer data from register R1 to register R2 and simultaneously perform an addition of the value stored in R1 with another register R3. How would the control word be structured to accomplish this? Explain how the control signals will be generated to enable the required operations (register selection, ALU operation, data transfer).

12. Imagine a customer adds multiple items to their shopping cart. How would you use a stack to manage the cart items efficiently, ensuring that the last item added is the first one removed when the customer proceeds to checkout? Explain PUSH and POP operations. Convert the following arithmetic expression into reverse polish notation (RPN) and show the stack operations for evaluating it.

a) 5 + 10 * 6 + (3 + 5) + 7 * (6 + 2)
b) 8 * 3 + (4 + 8) + (4 * 7 + 2) + 2 + 3
c) (A + B) * C - (D - E) * (F + G)

13. Write 14-bit control word for the following microoperations:

(a) R5 ← R5 – 1
(b) R6 ← Shl R1

14. Write a program to evaluate the following arithmetic statement by using three, two, one and zero address instructions:

(a) A-B+(C+D)/E+F/G+(J*K)
(b) (A*B+C) +D/E+F*G+I*J

15. A bus organization has eight 4-bit registers. Show, how multiplexers can be used to select registers in a common bus configuration?

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Assignment 1.3 (Module I)

Q1. You're a technical consultant for a startup that designs educational tools. Your team lead asks you to create a module explaining the fundamental architecture of a computer system to beginners. The goal is to first simplify the core components and then highlight how this architecture has evolved to meet today's demands.

The Challenge:

(i) Draw a block diagram showing the most basic, high-level architecture of a computer system. Label and explain the purpose of the three main functional units.

(ii) Considering the current computing landscape (smartphones, IoT devices, cloud computing), identify and briefly explain at least three significant changes or additions to this simple architecture that are now considered standard.

Q2. Convert the following numbers as per the question,

a) $(852.25)_8 = (?)_{16}$ b) $(325.05)_{16} = (?)_2$

c) $(111101.11001)_2 = (?)_8$ d) $(625.11)_{16} = (?)8$

e) $(5EE9.17)16 = (?)_8$ f) $(1100110110111)_2$ to hexadecimal

Q3. Convert the following numbers to Sign magnitude, 1's and 2's representation. Assume 8 bits will be used to store values.

a) $(+101)_{10}$ b) $(-99)_{10}$ c) $(-85)_{10}$ d) $(+115)_{10}$ e) $(-127)_{10}$

Q4. Perform the following operations in 2's complement considering 8-bit data. Also check the overflow situation.

a) $(-37)_{10} - (-13)_{10}$ b) $(+101)_{10} - (+11)_{10}$ c) $(+85)_{10} + (-34)_{10}$ d) $(-87)_{10} + (-12)_{10}$

Q5. Suppose you are a junior logic designer at a tech company. Your team has been tasked with creating a crucial component for a new processor, a 4-bit binary adder. The lead engineer has given you the following specifications and a scenario to test your understanding of the design process.

The Challenge:

A new client requires a simple processor to perform basic calculations. The core of this processor is an arithmetic logic unit (ALU), and the first step is to design a 4-bit binary adder to handle the addition of two 4-bit numbers, $A=A_3A_2A_1A_0$ and $B=B_3B_2B_1B_0$. The adder must also include a carry-in (C_{in}) and produce a 4-bit sum ($S=S_3S_2S_1S_0$) and a carry-out (C_{out}).

Q6. Imagine you are designing a reconfigurable 4-bit ALU for a low power microcontroller. The core function is a 4-bit binary adder/subtractor.

Your challenge: Design this circuit using four full-adder modules and a single control signal, M. When M=0, the circuit must perform addition (A+B). When M=1, it must perform subtraction (A−B). How would you implement this functionality? Draw the logic circuit diagram and explain the role of the M signal.

Q7. Your role is a digital logic designer at a company creating a specialized digital signal processor. The lead engineer has assigned you the task of implementing a fast multiplication unit using Booth's algorithm for signed 8-bit numbers in 2's complement representation.

The Challenge:

You need to demonstrate your understanding of the algorithm by walking through the multiplication of two numbers:

- **Multiplicand (M) = $(-24)_{10}$**
- **Multiplier (Q) = $(-19)_{10}$**

Q8. Suppose you are designing the bus system for a simple CPU. It has 8 registers, each holding 16 bits of data. To transfer data from any of these registers onto a common bus, you must design a multiplexer (MUX) system.

1. How many MUXs are required in total for this bus system?
2. What is the size of each individual MUX?
3. How many selection lines are needed to control these MUXs?

Q9. Imagine, you are a programmer debugging a financial application where floating- point calculations produce inconsistent results on different computers. Your team lead suspects a lack of standardized floating-point representation.

1. Explain why the IEEE 754 standard is crucial for solving this problem.
2. To demonstrate your understanding, convert the decimal number $(+81.25)10$ into its binary representation using the IEEE 754 standard.
3. Show the final 32-bit single-precision format.
4. Show the final 64-bit double-precision format.

Q10. Write a program to evaluate the arithmetic statement:

$$X = \frac{A - B + C * (D * E - F)}{G + H * K}$$

a) Using a general register computer with three address instructions
b) Using a general register computer with two address instructions
c) Using an accumulator type computer with one address instructions
d) Using a stack organizer computer with zero-address operation instructions.

Q11. Imagine you are an engineering student working on a team designing a high-speed processor for a new supercomputer. Your team has decided to use a ripple-carry adder for the processor's Arithmetic Logic Unit (ALU).

Explain the key significance of the carry-lookahead adder in the context of your team's project. How does it solve the problem of the ripple-carry adder, and why is this critical for a high-speed supercomputer?